

# UNIVERSITY INSTITUTE OF ENGINEERING

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BTECH CSE – CS 201

SEMESTER: 6TH

### CLOUD COMPUTING

(23CSH-307)

Unit No. 3 Chapter No. 1

Lecture No. 21

Topic : Introduction to Object, Block, and File storage types. Comparative analysis of storage use cases (e.g., S3 vs EBS vs EFS).

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# RECAP OF THE LAST LECTURE

- Deployment patterns are used to reduce risk during software releases.
- Blue-Green deployment enables simple switching and quick rollback.
- Canary deployment exposes changes to a small user group first.
- Rolling updates gradually replace old versions with new ones.
- Each pattern balances risk, cost, and availability differently.
- There is no universally best deployment strategy.
- Pattern selection depends on system requirements and constraints.
- Deployment strategies are essential for continuous delivery.
- These patterns are a core concept within DevOps practices.



# Learning Objectives

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- Fundamentals of Cloud Storage
- Object, Block, and File Storage
- Characteristics of Each Storage Type
- Comparative Use Case Analysis



# Fundamentals of Cloud Storage

- Cloud storage provides persistent data storage
- Data stored in distributed systems
- Accessible over network connections
- Abstracts physical storage hardware
- Designed for durability and availability
- Scales elastically with demand
- Managed by cloud providers
- Supports diverse application workloads

# Classification of Storage Types

- Storage classified by access method
- Object storage uses object-based access
- Block storage uses block-level access
- File storage uses hierarchical file access
- Each type optimized for specific needs
- Differ in performance and scalability
- Influence application architecture
- No one-size-fits-all solution



# Object Storage – Introduction

- Data stored as independent objects
- Each object contains data and metadata
- Objects identified by unique IDs
- Flat namespace without directories
- Accessed using REST-based APIs
- Designed for massive scalability
- Suitable for unstructured data
- High durability by design

# Object Storage – Characteristics

- Extremely scalable storage architecture
- Data replicated across multiple locations
- API-based access model
- Not mounted as a file system
- Optimized for large objects
- Higher latency compared to block storage
- Limited support for partial updates
- Cost-effective for large data volumes

# Block Storage – Introduction

- Provides raw storage blocks
- Attached directly to compute instances
- Requires file system formatting
- Acts like a virtual hard disk
- Low-latency access
- Suitable for transactional workloads
- Supports frequent read/write operations
- Commonly used for system disks





# Block Storage – Characteristics

- High performance and low latency
- Predictable IOPS behavior
- Strong consistency guarantees
- Limited scalability per volume
- Typically attached to one instance
- Requires manual volume management
- Supports snapshots for backup
- Ideal for databases and OS storage



# File Storage – Introduction

- Uses hierarchical directory structure
- Files organized into folders
- Shared file system access
- Mounted over network
- Familiar POSIX-style interface
- Enables concurrent access
- Suitable for shared workloads
- Simplifies legacy application migration

# File Storage – Characteristics

- Supports multiple readers and writers
- Centralized file storage system
- Moderate performance and latency
- Easier integration with applications
- Scales better than traditional NAS
- Suitable for shared application data
- Higher overhead than block storage
- Ideal for collaborative environments



# Overview of S3, EBS, and EFS

- S3 represents object storage
- EBS represents block storage
- EFS represents file storage
- Each optimized for different workloads
- Access mechanisms differ significantly
- Performance characteristics vary
- Scalability models are different
- Choice depends on application needs

# S3 vs EBS vs EFS – Access Model

- S3 accessed through REST APIs
- EBS attached directly to instances
- EFS mounted as network file system
- S3 is globally accessible
- EBS is instance-specific
- EFS supports shared access
- S3 not suitable for OS mounting
- EFS uses file-level operations

# S3 vs EBS vs EFS – Use Cases

- S3 used for backups and archival data
- S3 ideal for static website content
- EBS used for operating systems
- EBS suitable for databases
- EFS used for shared application storage
- EFS supports content management systems
- S3 supports big data storage
- EFS enables collaborative workloads



# Selecting Appropriate Storage

- Choose object storage for unstructured data
- Choose block storage for low-latency needs
- Choose file storage for shared access
- Consider access pattern requirements
- Evaluate performance sensitivity
- Assess scalability needs
- Match storage to application design
- Storage choice affects cost and efficiency

# Summary of the Lecture

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- Cloud provides multiple storage options
- Object, block, and file serve distinct roles
- Object storage excels in scalability
- Block storage excels in performance
- File storage excels in shared access
- Proper selection improves application efficiency
- Storage is a key architectural decision
- Understanding differences is essential





## Faculty-curated videos, NPTEL, Coursera, LinkedIn, or other relevant learning resources

- Cloud Computing: [https://onlinecourses.nptel.ac.in/noc21\\_cs14/preview](https://onlinecourses.nptel.ac.in/noc21_cs14/preview)
- Cloud Computing: [https://onlinecourses.nptel.ac.in/noc26\\_cs55/preview](https://onlinecourses.nptel.ac.in/noc26_cs55/preview)
- Google Cloud Computing Foundations: [https://onlinecourses.nptel.ac.in/noc20\\_cs55/preview](https://onlinecourses.nptel.ac.in/noc20_cs55/preview)
- Google Cloud Computing Foundations: [https://onlinecourses.nptel.ac.in/noc24\\_cs131/preview](https://onlinecourses.nptel.ac.in/noc24_cs131/preview)

# Next Lecture

Understanding storage classes (Standard, Infrequent Access, Archive), defining lifecycle policies for cost optimization.

# Quiz/ FAQ's

1. What is the primary difference between object and block storage?

Object storage manages data as objects accessed via APIs, while block storage provides raw disk blocks attached to compute instances.

2. Why is object storage suitable for large unstructured data?

Because it scales massively and stores data with metadata without requiring a file system.

3. When is block storage preferred?

Block storage is preferred for performance-sensitive workloads like databases and operating systems.

4. Why is file storage useful in cloud environments?

It allows multiple systems to access shared files using a familiar directory structure.

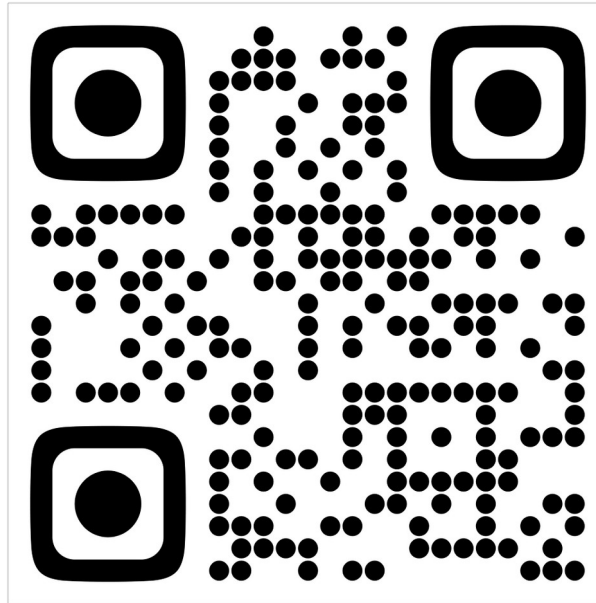
5. How do S3, EBS, and EFS differ in purpose?

S3 is for object data, EBS is for block-level storage, and EFS is for shared file storage.

# References/ Articles/ Videos

- <https://aws.amazon.com/s3/>
- <https://aws.amazon.com/ebs/>
- <https://aws.amazon.com/efs/>
- <https://www.ibm.com/topics/block-storage>
- <https://cloud.google.com/architecture/storage-options>

# Class-Wise Feedback



# Thank You

For queries

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